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Aggregation and Random Forests

Machine Learning
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Outline

- Introduction
- Bagging
- Random Forest
- Variables importance
- Implementation
- Conclusion

Introduction

- We present in this chapter algorithms based on a random construction of a family of models : bagging for bootstrap aggregating (Breiman 1996) and the random forests of Breiman (2001) which proposes an improvement of *bagging* specific to models defined by binary trees (CART).
- The principle of *bagging* applies to any modeling method (linear regression, CART, neural networks..) but are mostly interesting, and significantly reduces the prediction error, only in the case of *unstable* models.
- It is mainly implemented in association with binary trees as a basic models.
- The already underlined instability of trees appears as a property favoring the reduction of variance by aggregation of these models.

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Principle and algorithm

- Let Y be a quantitative or qualitative variable, $X^1, \ldots, X^p$ the explanatory variables, $\hat{f}(\mathbf{x})$ a predictor, with $\mathbf{x} = (x^1, \ldots, x^p) \in \mathbb{R}^p$.
- We denote by n the number of observations and

$$\mathbf{Z} = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)\}$$

  a sample with distribution F.
- Considering B *independent samples* denoted $\{\mathbf{Z}_b\}_{b=1,\ldots B}$, a predictor by model aggregation is defined below in the case where the variable to explain Y is :
  - quantitative : $\widehat{f}_B(.) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}_{\mathbf{Z}_b}(.)$ (mean)
  - qualitative : $\widehat{f}_B(.) = \arg\max_j \operatorname{card}\left\{b \mid \widehat{f}_{\mathbf{Z}_b}(.) = j\right\}$ (majority vote)

Principle and algorithm

- The principle is elementary, averaging the predictions of several independent models allows to reduce the variance and therefore to reduce the prediction error.
- However, it is unrealistic to consider B independent samples. This would require too much data.
- These samples are therefore replaced by B bootstrap samples each obtained by n draws with replacement according to the empirical measure $\widehat{F}_n$.
- This leads to the following algorithm.

Bagging - Algorithm

Bagging - Algorithm

- Let $\mathbf{x}_0$
- Let $\mathbf{Z} = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)\}$ a learning sample
- For $b = 1$ to B
  - Draw a bootstrap sample of size n with replacement $\mathbf{z}_b$.
  - Estimate $\widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)$ with the bootstrap sample.
- Compute the mean $\widehat{f}_B(\mathbf{x}_0) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)$ or the result of a majority vote.

Bagging - Principle

Bootstrap AGGregatING

- Variance reduction : by aggregating independent predictions
  - Aggregation : average (regression), majority vote (classification)
- Bootstrap trick : get new data from themselves by resampling !
  - Caution : new data remain (slightly) dependent on the initial ones

Bagging - Introductive example

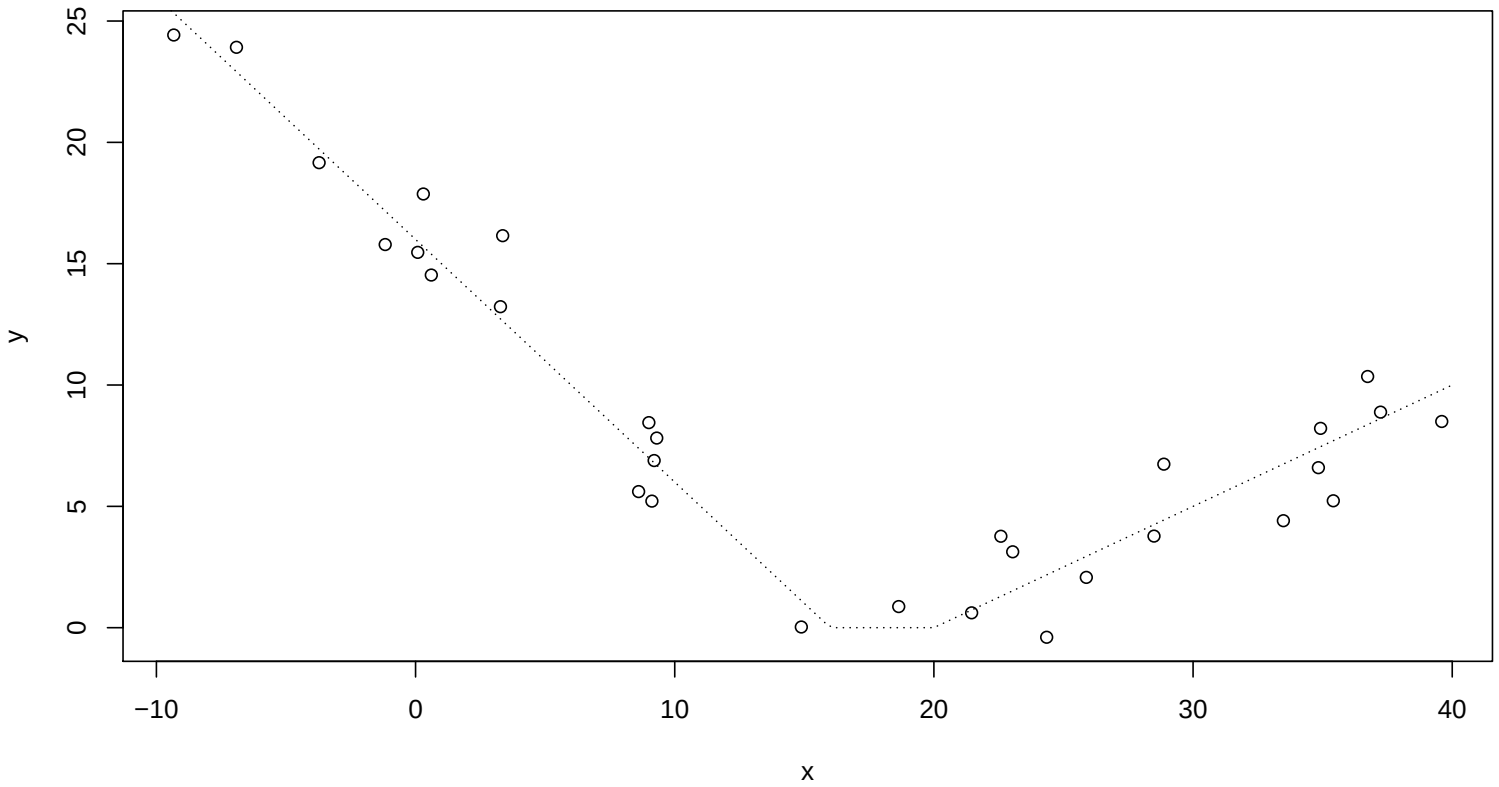

Figure – Original data

Bagging - Introductive example

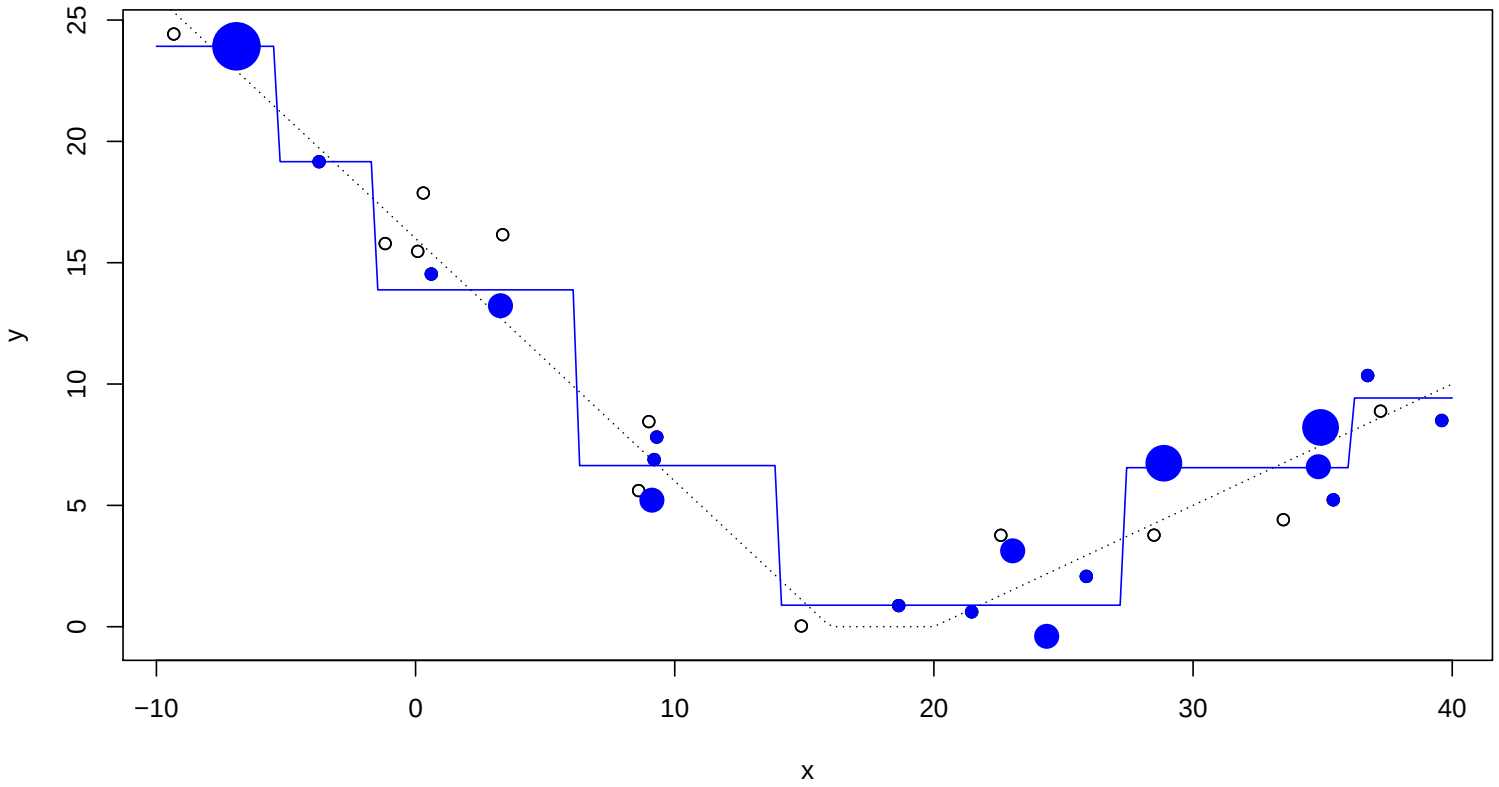

Figure – Bootstrap sample $n^o 1$ (in blue), and corresp. prediction with tree. The point size is proportional to the number of replicates.

Bagging - Introductive example

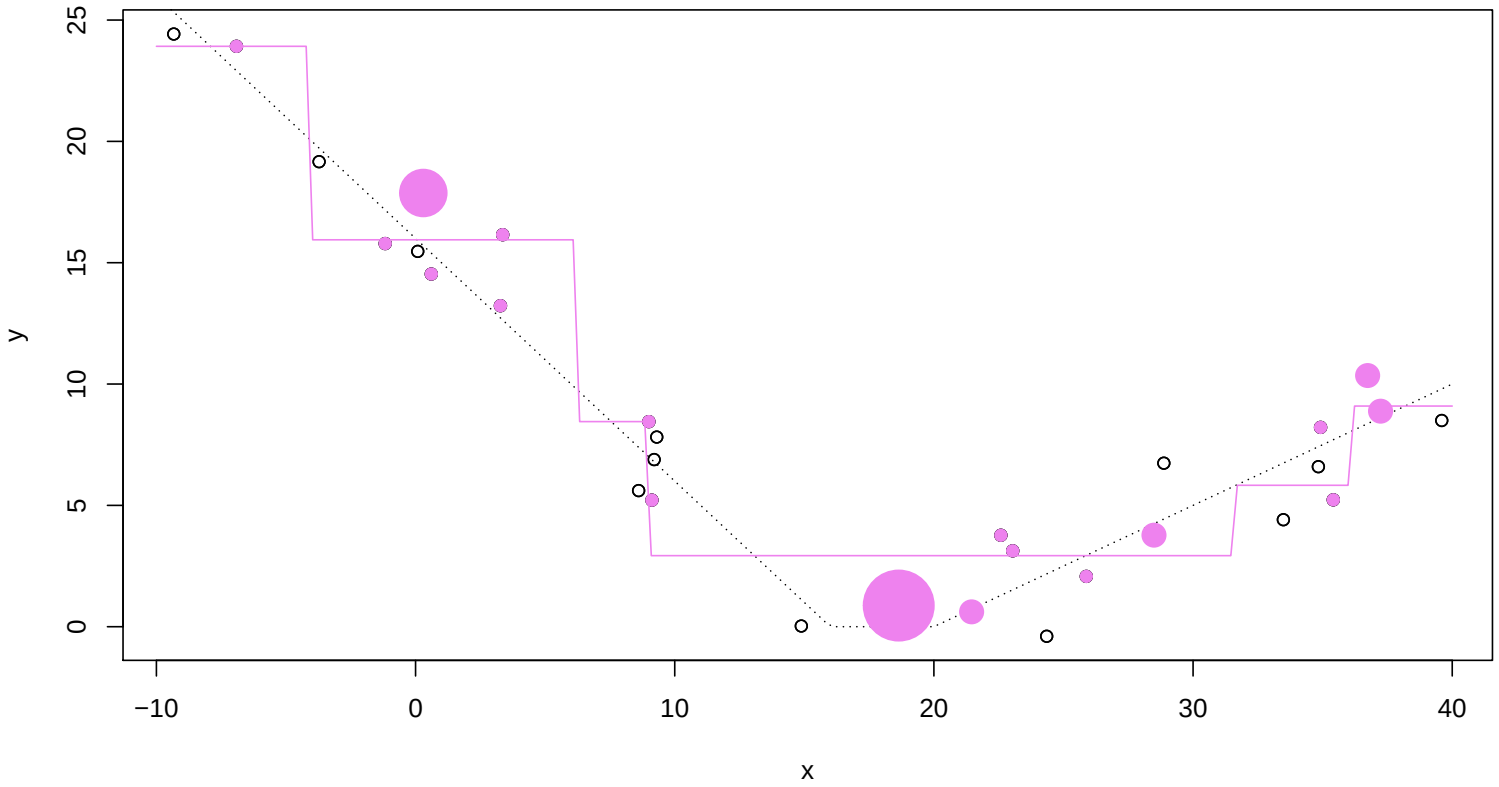

Figure – Bootstrap sample $n^o 2$ (in violet), and corresp. prediction with tree. The point size is proportional to the number of replicates.

Bagging - Introductive example

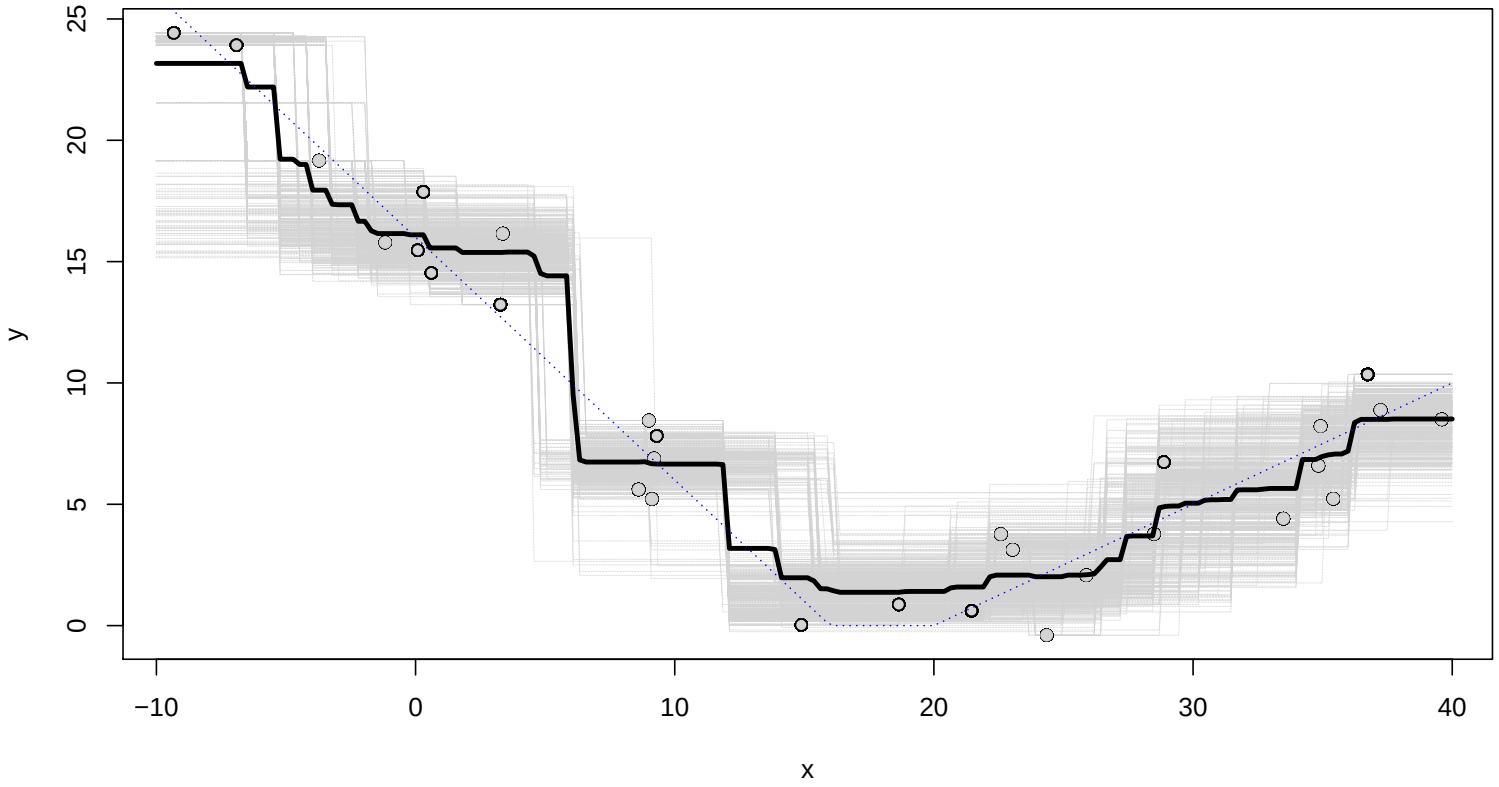

Figure – 500 bootstrap samples (grey), corresp. predictions with tree, and their average (bold line).

Bagging - Theory

- The B boostrap samples are built on the same learning sample $\mathbf{Z} = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)\}$ and therefore the estimators $\widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)$ are not independent.
- We assume that,
  - $\forall b$, $\mathbb{E}(\widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)) = f(\mathbf{x}_0)$,
  - $\forall b$, $\mathrm{Var}(\widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)) = V(\mathbf{x}_0)$
  - $\forall b \neq b'$, $\mathrm{Corr}(\widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0), \widehat{f}_{\mathbf{z}_{b'}}(\mathbf{x}_0)) = \rho(x_0)$.
- Then, in the regression case, we obtain

$$\mathbb{E}(\widehat{f}_B(\mathbf{x}_0)) = f(\mathbf{x}_0)$$

$$\mathrm{Var}(\widehat{f}_B(\mathbf{x}_0)) = \rho(x_0)V(\mathbf{x}_0) + \frac{(1-\rho(x_0))}{B}V(\mathbf{x}_0)$$

Bagging - Theory

$$\begin{aligned}\mathrm{Var}(\widehat{f}_B(\mathbf{x}_0)) &= \rho(x_0)V(\mathbf{x}_0) + \frac{(1-\rho(x_0))}{B}V(\mathbf{x}_0) \\ &\to \rho(x_0)V(\mathbf{x}_0) \text{ as } B \to +\infty\end{aligned}$$

- If the correlation term $\rho(x_0)$ is small, the variance of the aggregated predictor $\widehat{f}_B(\mathbf{x}_0)$ is much smaller than the one of a single predictor.
- This underlines the importance of finding low correlated predictors $(\widehat{f}_b(\mathbf{x}_0))_{1\leq b\leq B}$, which is at the core of the **Random forests** algorithm.

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- In the specific case of binary decision tree models (CART), Breiman (2001) proposes an improvement of the *bagging* by adding a random component.
- The objective is to make the aggregated trees more independent by adding randomness in the choice of the variables which are involved in the prediction.
- Since the initial publication of the algorithm, this method has been widely tested and compared with other procedures, it becomes in many machine learning articles the method to beat in terms of prediction accuracy.
- Theoretical convergence properties, difficult to study, have been published quite recently (Scornet et al. 2015).
- However, when the underlying problem is linear, it can also lead to bad results.

Algorithm

The bagging is applied to binary decision trees by adding a random selection of m explanatory variables among the p variables.

Random Forests algorithm

- Let $\mathbf{x}_0$
- Let $\mathbf{Z} = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_n, Y_n)\}$ a learning sample
- For $b = 1$ to B
  - Take a bootstrap sample $\mathbf{z}_b$
  - Estimate a tree on this sample with **randomization** of the variables : the search for each optimal division is preceded by a random selection of a subset of m predictors.
- Calculate the mean estimate $\widehat{f}_B(\mathbf{x}_0) = \frac{1}{B}\sum_{b=1}^{B} \widehat{f}_{\mathbf{z}_b}(\mathbf{x}_0)$ or the result of a majority vote.

Parameters of the algorithm

- The pruning strategy can, in the case of random forests, be quite elementary.
- Indeed, pruned trees may be strongly correlated because they may involve the same variables appearing to be the most explanatory.
- In the default strategy of the algorithm, it is simply the minimum number of observations per leaf which limits the size of the tree, it is set to 5 by default.
- We therefore aggregate rather complete trees, which are considered of low bias but of high variance.

Parameters of the algorithm

- The random selection of a reduced number of m potential predictors at each stage of the construction of the trees significantly increases the variability by highlighting other variables.
- Each tree is obviously less efficient, sub-optimal, but, united being strength, aggregation ultimately leads to good results.
- The number m of variables drawn randomly can, according to the examples, be a sensitive parameter with default choices are not always optimal :
  - $m = \sqrt{p}$ in a classification problem,
  - $m = p/3$ in a regression problem.
- The iterative evaluation of the *out-of-bag* error makes it possible to control the number B of trees in the forest as well as to optimize the choice of m. It is nevertheless a cross-validation procedure which is preferably used to optimize m.

Bagging - Out-Of-Bag data

Out-Of-Bag (OOB) data

For each bootstrap sample :

- Let $U_1^\star, \dots, U_N^\star$ be random variables representing the boostrapped indices. The probability that a given data z_i is not chosen is :

$$\mathbb{P}\left(z_{U_1^\star} \neq z_i, \dots, z_{U_N^\star} \neq z_i\right) = \left(1 - \frac{1}{N}\right)^N \underset{N \to +\infty}{\longrightarrow} e^{-1} \approx 0.367$$

- The non-chosen data are called Out-Of-Bag (OOB). They can be used as a test set inside the bootstrap loop

The OOB error is obtained by averaging prediction errors over OOB data

Bagging - Out-Of-Bag data

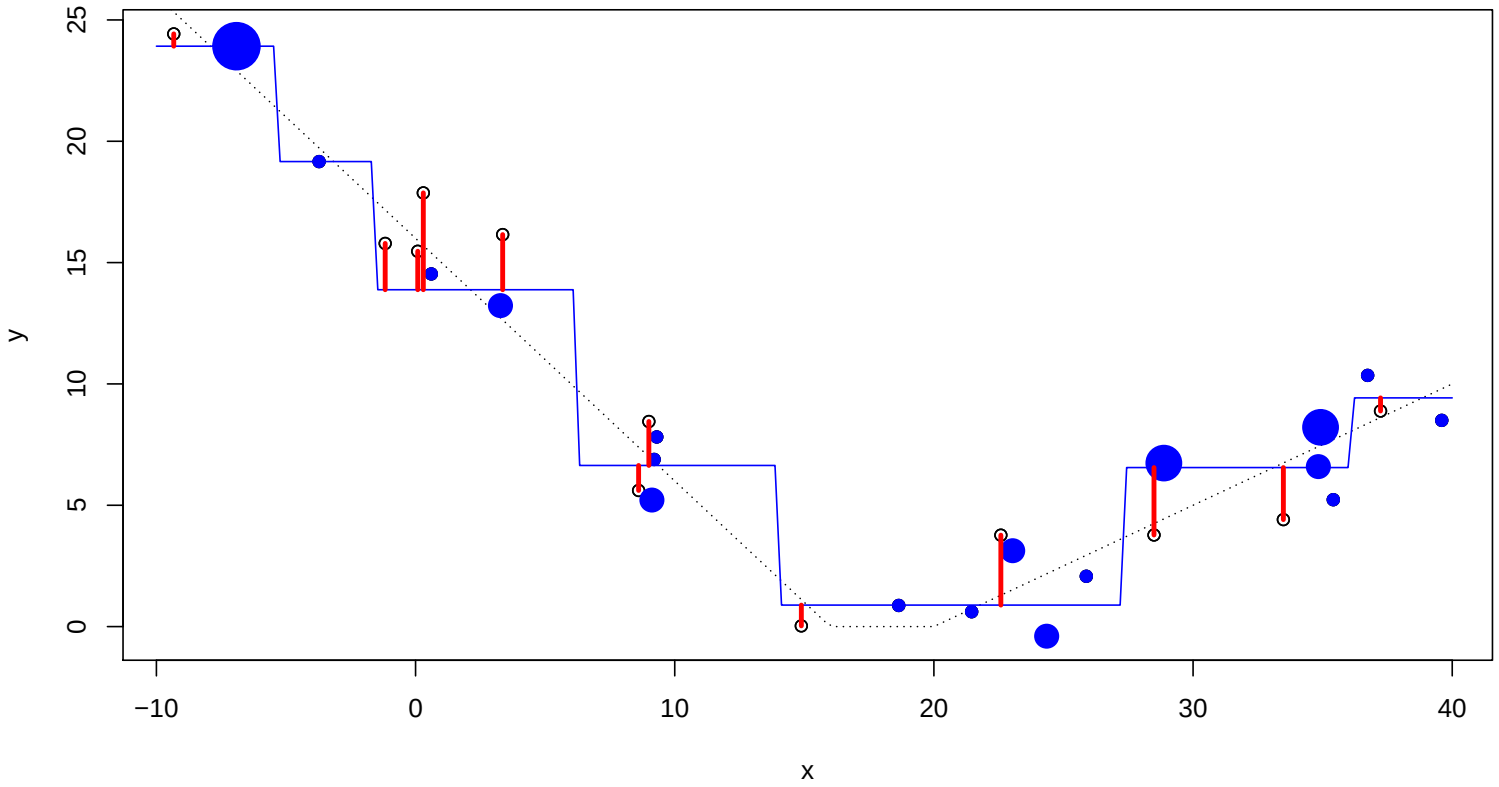

Figure – Residuals for the OOB bootstrap sample $n^o 1$ (red bars).

Bagging - Out-Of-Bag data

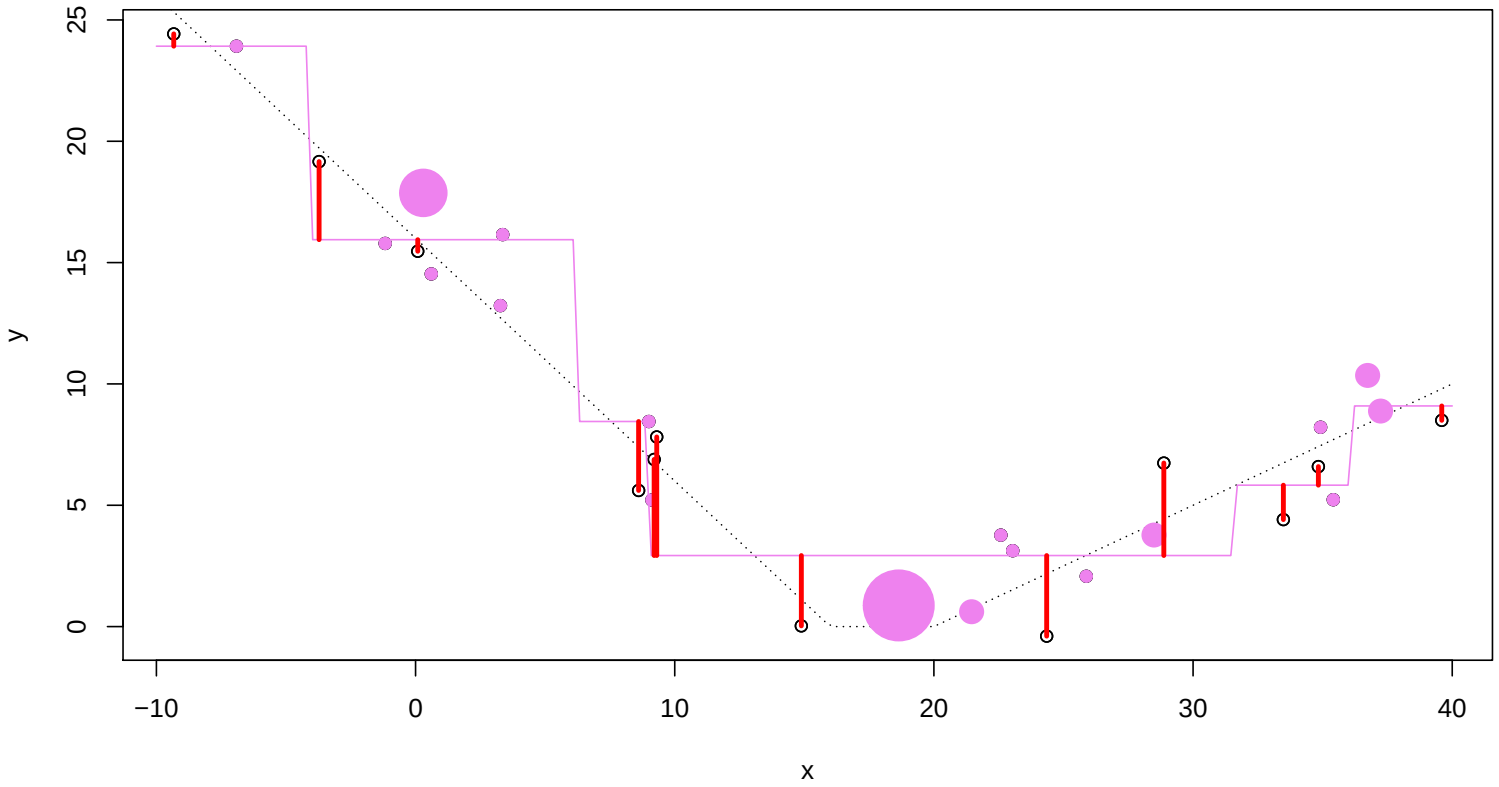

Figure – Residuals for the OOB bootstrap sample $n^o 2$ (red bars).

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Variables importance

- Random forests generally have a good accuracy, they are easily implementable, parallelisable but not easy to interpret like any model built by aggregation, leading to a black-box model.
- To favor interpretation, indexes of importance for each explanatory variable have been introduced.
- This is obviously all the more useful as the variables are very numerous.
- Two criteria have been proposed to evaluate the importance of the variable X^j.

Interpretation - Variable importance

How can we quantify the importance of a variable X^j in random forest ?

Mean decrease Impurity : MDI (Breiman, 2003)

- Average the decrease of heterogeneity when X^j is chosen as a split.

Mean Decrease Accuracy : MDA (Breiman, 2001)

- Based on a permutation of variables :
  - Compute the OOB error for the subsample of OOB data.
  - Compare with the OOB error when permuting at random the values of the variable X^j in the learning sample (but keeping the output Y unchanged).

Mean decrease Impurity

More formally, with previous notations, the MDI of the variable X^j is defined by

$$MDI(X^j) = \frac{1}{Bn} \sum_{b=1}^{B} \sum_{\kappa \in \mathcal{T}_b, j_\kappa^* = j} \left[D_\kappa - \left(D_{\kappa_L}(t_\kappa^*, j_\kappa^*) + D_{\kappa_R}(t_\kappa^*, j_\kappa^*) \right) \right],$$

- $\{\mathcal{T}_b, 1 \leq b \leq B\}$ is the collection of trees in the forest,
- (t_κ^*, j_κ^*) the split retained at node κ :
  - j_κ^* corresponds to the optimal variable selected for the split
  - t_κ^* corresponds to the optimal threshold along the j_κ^* variable.

Mean decrease Accuracy

- Consider a variable X^j and denote by $\mathcal{D}_{b,n}$ the out-of-bag data set of the b-th tree and $\mathcal{D}^j_{b,n}$ the same data set where the values of X^j have been randomly permuted.
- Denote by $\widehat{f}_{\mathbf{z}_b}$ the b-th tree estimate and

$$R_n\left[\widehat{f}_{\mathbf{z}_b}, \mathcal{D}\right] = \frac{1}{|\mathcal{D}|} \sum_{i,(X_i,Y_i)\in\mathcal{D}} (Y_i - \widehat{f}_{\mathbf{z}_b}(X_i))^2.$$

- The MDA is defined by

$$MDA(X^j) = \frac{1}{B}\sum_{b=1}^{B}\left\{R_n\left[\widehat{f}_{\mathbf{z}_b}, \mathcal{D}^j_{b,n}\right] - R_n\left[\widehat{f}_{\mathbf{z}_b}, \mathcal{D}_{b,n}\right]\right\}.$$

Example on ozone data

```
> library(randomForest)
> rf.reg <- randomForest(O3obs ., data = datappr, xtest = datestr[, -2],
   ytest = datestr[, "O3obs"], ntree = 500, do.trace = 50, importance = TRUE)
```

	Out -of	- bag	Test	set
Tree	MSE	%Var(y)	MSE	%Var(y)
50	697.9	40.77	568.5	36.75
100	689.5	40.28	555.9	35.93
150	683.8	39.95	563.2	36.41
200	685.4	40.04	561	36.27
250	678.2	39.62	564.2	36.47
300	675.1	39.44	569.2	36.79
350	676.8	39.54	572.8	37.02
400	674.3	39.39	571.4	36.93
450	673.9	39.37	571.5	36.94
500	674.3	39.39	569.6	36.82

```
> round(importance(rf.reg), 2)

          %IncMSE   IncNodePurity
JOUR        1.98       11011.79
MOCAGE     41.46      388657.27
TEMPE      51.73      409018.57
STATION    21.73       75350.42
VentMOD    12.95       91387.20
VentANG    18.81      124908.37
SRMH2O     16.76      114463.05
LNO2        7.73       84152.34
LNO        10.04       74387.32
```

Details (from R help file of function `importance`)

"The first measure [%IncMSE] is computed from permuting OOB data : For each tree, the prediction error on the out-of-bag portion of the data is recorded (error rate for classification, MSE for regression). Then the same is done after permuting each predictor variable. The difference between the two are then averaged over all trees, and normalized by the standard deviation of the differences.

The second measure [IncNodePurity] is the total decrease in node impurities from splitting on the variable, averaged over all trees. For classification, the node impurity is measured by the Gini index. For regression, it is measured by residual sum of squares."

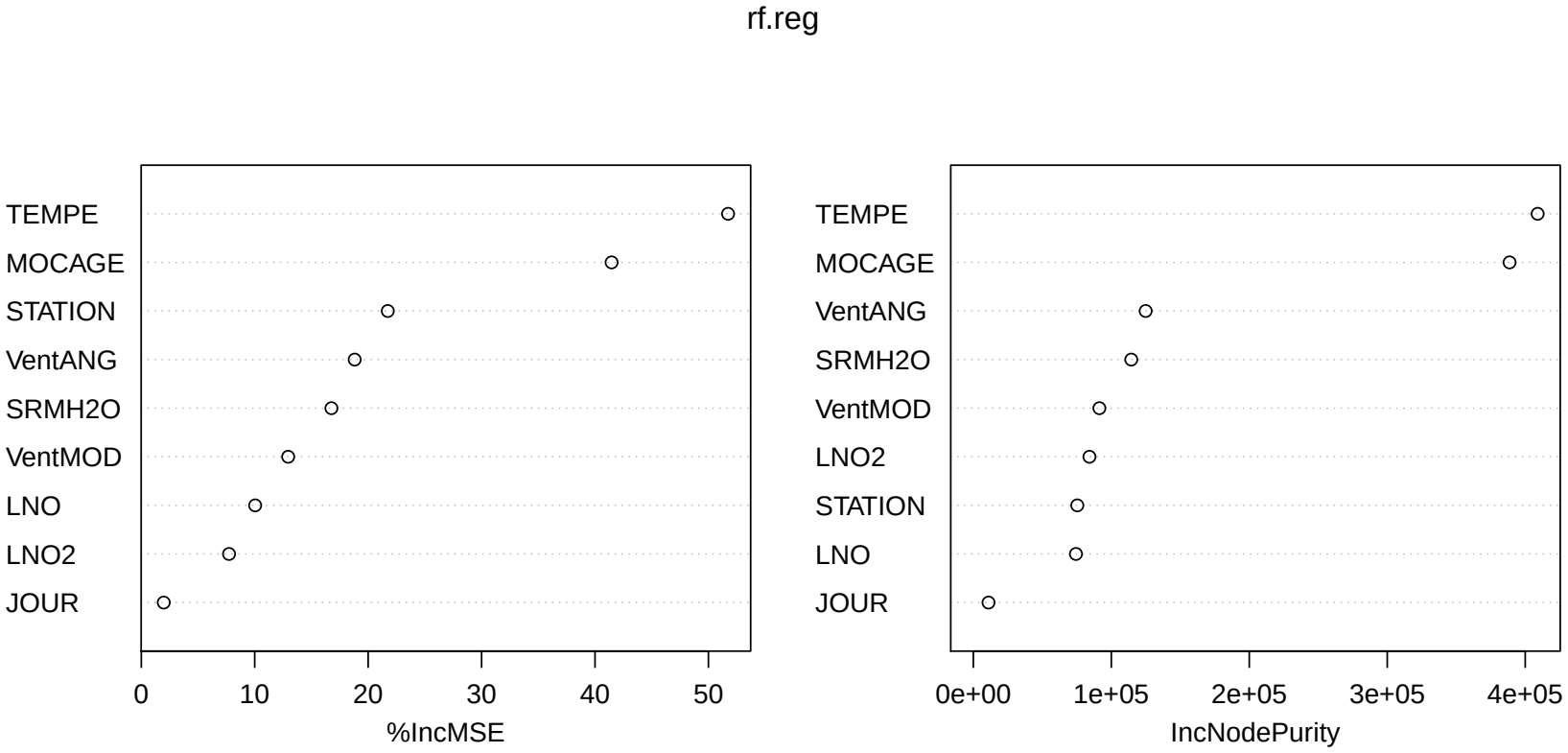

Figure – Variable importance plot, returned by the R function `importance`

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Implementation

- The `randomForest` library of R interfaces the original program developed in Fortran77 by Leo Breiman and Adele Cutler which maintains the site dedicated to this algorithm https ://math.usu.edu/adele/forests/
- A very efficient and close version of the original algorithm is available in the `Scikit-learn` library of Python.
- Another version suitable for big data is available in the `MLlib` library of Spark, a technology developed to interface different hardware/software architectures with distributed data file management systems (Hadoop).

Implementation

- In addition to the usual parameters : number of trees, maximum depth of trees, and number of variables drawn at random to build a subdivision at each node, this implementation adds two parameters : `subsamplingRate` and `maxBins`, which have a default value.
- These parameters play an important role, in drastically reducing the computation time, but, on the other hand, in restricting the precision of the estimate.
- They regulate the balance between computation time and precision of the estimate as a subsampling in the data would do.

Implementation

- **subsamplingRate =1.0** subsamples before building each tree. With the default value, it is the classic version of random forests with B Bootstrap samples of size n but if this rate is less than 1, smaller samples are drawn. The sample are then more distinct (or independent) for each tree. The variance is therefore reduced (more independent trees) but the bias increases because each tree is built with a smaller data set.
- **maxBins = 32** is the maximum number of categories that are considered for a qualitative variable. Only the most frequent modalities of a qualitative variable are taken into account, the others are automatically grouped into a `other` modality. As previously, the time to determine a better division is obviously largely influenced by the number of modalities. Reducing the number of possible values is finally another way of reducing the computation time but it would be appropriate to guide the groupings of the modalities to avoid misinterpretations.

Conclusion

Random Forests are used for different purposes (see the dedicated site : https ://math.usu.edu/adele/forests/)

- Similarity or proximity between observations : after building each tree, increment by 1 the similarity or proximity of two observations that are in the same leaf. Sum on the trees of the forest, normalize by the number of trees. We can compute a matrix of dissimilarities that results from them.
- Detection of multidimensional atypical observations : outliers or novelties that correspond to observations which do not belong to known classes. A criterion of "abnormality" with respect to a class is based on the previous notion of proximities of an observation to the other observations of its class.
- Another algorithm, inspired by Random Forests has been developed for anomaly detection, it is called isolation forest.
- Random forests are used for the imputation of missing data.
- Adaptations to take into account censored data to model survival times correspond to the survival forest algorithm.

References

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